```
> with(Student[LinearAlgebra]) : with(LinearAlgebra) : with(linalg)
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[*BlockDiagonal, GramSchmidt, JordanBlock, LUdecomp, QRdecomp, Wronskian, addcol, addrow, adj, adjoint, angle, augment, backsub, band, basis, bezout, blockmatrix, charmat, charpoly, cholesky, col, coldim, colspace, colspan, companion, concat, cond, copyinto, crossprod, curl, definite, delcols, delrows, det, diag, diverge, dotprod, eigenvals, eigenvalues, eigenvectors, eigenvects, entermatrix, equal, exponential, extend, ffgausselim, fibonacci, forwardsub, frobenius, gausselim, gaussjord, geneqns, genmatrix, grad, hadamard, hermite, hessian, hilbert, htranspose, ihermite, indexfunc, innerprod, intbasis, inverse, ismith, issimilar, iszero, jacobian, jordan, kernel, laplacian, leastsqrs, linsolve, matadd, matrix, minor, minpoly, mulcol, mulrow, multiply, norm, normalize, nullspace, orthog, permanent, pivot, potential, randmatrix, randvector, rank, ratform, row, rowdim, rowspace, rowspan, rref, scalarmul, singularvals, smith, stackmatrix, submatrix, subvector, sumbasis, swapcol, swaprow, sylvester, toeplitz, trace, transpose, vandermonde, vecpotent, vectdim, vector, wronskian*] **(1)**

```
> A := Matrix([[0,-2, 0], [1,-2, 0], [0, 0,-2]])
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$$A := \begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad (2)$$

```
> u1 := Vector([0, 0, 1])
```

$$u1 := \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (3)$$

```
> lambda1 := -2
```

$$\lambda 1 := -2 \quad (4)$$

```
> A . u1 - lambda1 . u1
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (5)$$

```
> u2 := Vector([1 + I, 1, 0])
```

$$u2 := \begin{bmatrix} 1+\mathrm{I} \\ 1 \\ 0 \end{bmatrix} \quad (6)$$

> $lambda2 := -1 + I$

$$\lambda 2 := -1 + \mathrm{I} \quad (7)$$

> $A \cdot u2 - lambda2 \cdot u2$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

> $u3 := Vector([1 - I, 1, 0])$

$$u3 := \begin{bmatrix} 1-\mathrm{I} \\ 1 \\ 0 \end{bmatrix} \quad (9)$$

> $lambda3 := -1 - I$

$$\lambda 3 := -1 - \mathrm{I} \quad (10)$$

> $A \cdot u3 - lambda3 \cdot u3$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

> $P := Matrix([\langle u1 \rangle, \langle u2 \rangle, \langle u3 \rangle])$

$$P := \begin{bmatrix} 0 & 1+\mathrm{I} & 1-\mathrm{I} \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (12)$$

> $J := DiagonalMatrix([lambda1, lambda2, lambda3])$

$$J := \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1+\mathrm{I} & 0 \\ 0 & 0 & -1-\mathrm{I} \end{bmatrix} \quad (13)$$

> $A - P \cdot J \cdot inverse(P)$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (14)$$

> $MatrixExponential(t \cdot J)$

$$\begin{bmatrix} e^{-2t} & 0 & 0 \\ 0 & e^{-t}\cos(t) + I\,e^{-t}\sin(t) & 0 \\ 0 & 0 & e^{-t}\cos(t) - I\,e^{-t}\sin(t) \end{bmatrix} \quad (15)$$

> $expA := MatrixExponential(t \cdot A)$

$$expA := \begin{bmatrix} e^{-t}\cos(t) + e^{-t}\sin(t) & -2\,e^{-t}\sin(t) & 0 \\ e^{-t}\sin(t) & e^{-t}\cos(t) - e^{-t}\sin(t) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \quad (16)$$

> $Map(limit, expA, t = \text{infinity})$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (17)$$

> $P := Matrix([[1, 0, 0, 1], [0, 1, 0, 1], [0, 0, 1, 0], [1, 1, 1, 1]])$

$$P := \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad (18)$$

> $det(P)$

$$-1 \quad (19)$$

> $J := DiagonalMatrix([2, 2, -1, 0])$

(20)

$$J := \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (20)$$

> $A := P \cdot J \cdot inverse(P)$

$$A := \begin{bmatrix} 0 & -2 & -2 & 2 \\ -2 & 0 & -2 & 2 \\ 0 & 0 & -1 & 0 \\ -2 & -2 & -5 & 4 \end{bmatrix} \quad (21)$$

> $det(A)$

$$0 \quad (22)$$

> $CharacteristicPolynomial(A, r)$

$$r^4 - 3\,r^3 + 4\,r \quad (23)$$

> $Eigenvectors(A)$

$$\begin{bmatrix} 0 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \quad (24)$$

> $JordanForm(A)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \quad (25)$$

> $A := Matrix([[2, 1, 0], [0, 2, 1], [0, 0, 2]])$

$$A := \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad (26)$$

> $Eigenvalues(A)$

$$\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \tag{27}$$

> $JordanForm(A)$

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \tag{28}$$

> $MatrixExponential(t \cdot A)$

$$\begin{bmatrix} e^{2t} & t e^{2t} & \frac{t^2 e^{2t}}{2} \\ 0 & e^{2t} & t e^{2t} \\ 0 & 0 & e^{2t} \end{bmatrix} \tag{29}$$

> $X := Vector([x, y, z])$

$$X := \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{30}$$

> $Xsol := simplify(MatrixExponential(t \cdot A) \cdot X)$

$$Xsol := \begin{bmatrix} e^{2t}\left(x + y t + \frac{1}{2} z t^2\right) \\ e^{2t}(z t + y) \\ e^{2t} z \end{bmatrix} \tag{31}$$

> $A := Matrix([[0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1], [-1, 0, -2, 0]])$

$$A := \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -2 & 0 \end{bmatrix} \tag{32}$$

> $JordanForm(A)$

$$\begin{bmatrix} -\mathrm{I} & 1 & 0 & 0 \\ 0 & -\mathrm{I} & 0 & 0 \\ 0 & 0 & \mathrm{I} & 1 \\ 0 & 0 & 0 & \mathrm{I} \end{bmatrix} \tag{33}$$

> $MatrixExponential(t \cdot A)$

$$\begin{bmatrix} \cos(t) + \frac{\sin(t)\,t}{2} & -\frac{t\cos(t)}{2} + \frac{3\sin(t)}{2} & -\frac{s}{} \cdots \\ \frac{t\cos(t)}{2} - \frac{\sin(t)}{2} & \cos(t) + \frac{\sin(t)\,t}{2} & \frac{t\cos(}{2} \cdots \\ -\frac{\sin(t)\,t}{2} & \frac{t\cos(t)}{2} - \frac{\sin(t)}{2} & \cos(t) \cdots \\ -\frac{t\cos(t)}{2} - \frac{\sin(t)}{2} & -\frac{\sin(t)\,t}{2} & -\frac{t\cos(}{2} \cdots \end{bmatrix} \tag{34}$$

> $X := Vector([w, x, y, z])$

$$X := \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} \tag{35}$$

> $Xsol := simplify(MatrixExponential(t \cdot A) \cdot X)$

$Xsol :=$ (36)

$$\begin{bmatrix} \frac{((w+y)\,t+3\,x+z)\sin(t)}{2} - \frac{((x+z)\,t-2\,w)\cos(t)}{2} \\ \frac{((x+z)\,t-w+y)\sin(t)}{2} + \frac{((w+y)\,t+2\,x)\cos(t)}{2} \\ \frac{((-w-y)\,t-x+z)\sin(t)}{2} + \frac{((x+z)\,t+2\,y)\cos(t)}{2} \\ \frac{((-x-z)\,t-w-3\,y)\sin(t)}{2} - \frac{((w+y)\,t-2\,z)\cos(t)}{2} \end{bmatrix}$$

> $solve(1 - x\hat{}2 = 0, x)$

(37)

$$-1, 1 \tag{37}$$

> $eq4 := diff(x(t), t) = 1 - x(t)^2$

$$eq4 := \frac{d}{dt} x(t) = 1 - x(t)^2 \tag{38}$$

> $sol41 := dsolve(\{eq4, x(0) = -1\}, x(t))$

$$sol41 := x(t) = -1 \tag{39}$$

> $sol42 := dsolve(\{eq4, x(0) = 1\}, x(t))$

$$sol42 := x(t) = 1 \tag{40}$$

> $sol43 := dsolve(\{eq4, x(0) = -2\}, x(t))$

$$sol43 := x(t) = \coth\left(-\operatorname{arctanh}\left(\frac{1}{2}\right) + t\right) \tag{41}$$

> $sol44 := dsolve(\{eq4, x(0) = 0\}, x(t))$

$$sol44 := x(t) = \tanh(t) \tag{42}$$

> $sol45 := dsolve(\{eq4, x(0) = 2\}, x(t))$

$$sol45 := x(t) = \coth\left(\operatorname{arctanh}\left(\frac{1}{2}\right) + t\right) \tag{43}$$

> $sol43 := convert(convert(sol43, \exp), \exp)$

$$sol43 := x(t) = \frac{e^{2t} + 3}{e^{2t} - 3} \tag{44}$$

> $sol44 := convert(convert(sol44, \exp), \exp)$

$$sol44 := x(t) = \frac{e^{2t} - 1}{e^{2t} + 1} \tag{45}$$

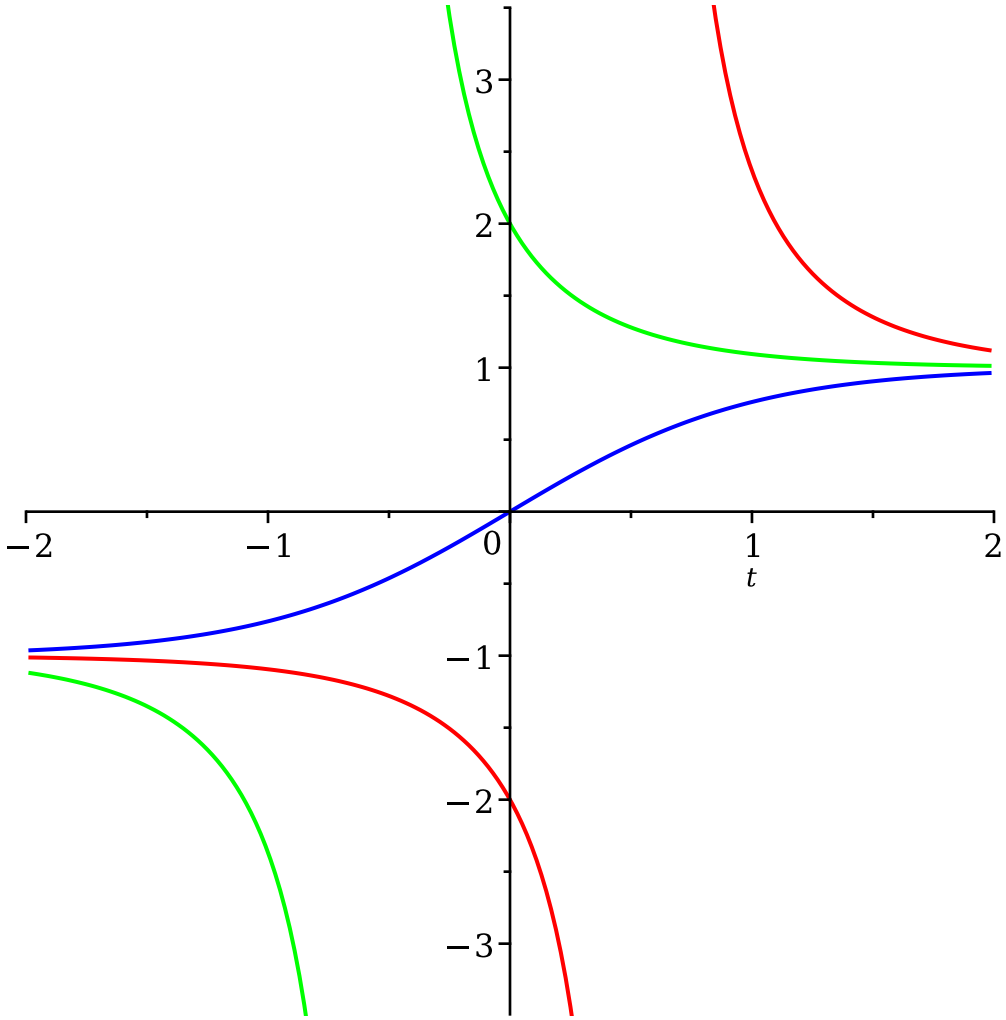
> $sol45 := convert(convert(sol45, \exp), \exp)$

$$sol45 := x(t) = \frac{3e^{2t} + 1}{3e^{2t} - 1} \tag{46}$$

> $with(plots)$

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*] (47)

> $plot([rhs(sol43), rhs(sol44), rhs(sol45)], t = -2..2, color = [red, blue, green])$




```
> solve(denom(rhs(sol43)), t)
```

$$\frac{\ln(3)}{2} \quad (48)$$

```
> solve(denom(rhs(sol45)), t)
```

$$-\frac{\ln(3)}{2} \quad (49)$$

```
> limit(rhs(sol43), t = -infinity)
```

$$-1 \quad (50)$$

```
> limit(rhs(sol44), t = infinity)
```

$$1 \quad (51)$$

```
> limit(rhs(sol44), t = -infinity)
```

$$-1 \quad (52)$$

```
> limit(rhs(sol45), t = infinity)
```

$$1 \quad (53)$$

```
>
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